FRAMEWORK ARCHITECTURE • PART 01

Performance Marketing Formula & Diagnostic Framework

Master mathematical dependencies, business unit economics, and operational ad channel flows.

Core Performance Metrics

FOUNDATION

```
CTR = (Clicks ÷ Impressions) × 100
CPC = Spend ÷ Clicks
CVR = (Conversions ÷ Clicks) × 100
CPA = Spend ÷ Conversions
CPL = Spend ÷ Leads
CPM = (Spend ÷ Impressions) × 1,000
ROAS = Revenue ÷ Spend
AOV = Revenue ÷ Orders
```

Key Inter-Metric Relationships

THE POWER RATIOS

```
CPA = CPC ÷ CVR
ROAS = AOV ÷ CPA
```

- CPA is strictly governed by incoming traffic cost (CPC) vs on-site efficiency (CVR).
- ROAS scales by elevating basket size (AOV) or reducing acquisition cost (CPA).

Meta Specific Metrics

AUCTION & ADS

```
Frequency = Impressions ÷ Reach
Reach = Impressions ÷ Frequency
Link CTR = (Link Clicks ÷ Impressions) × 100
Outbound CTR = (Outbound Clicks ÷ Impressions) × 100
Cost per Result = Spend ÷ Results
```

Business Economics & Bottom-Line Profitability

FINANCIAL VIABILITY

```
COGS = Cost of Goods Sold
Gross Profit = Revenue − COGS
Gross Margin % = (Gross Profit ÷ Revenue) × 100
Break-even ROAS = 1 ÷ Gross Margin %
Contribution Profit = Revenue − COGS − Ad Spend − Variable Costs
```

Actual ROAS must strictly exceed Break-even ROAS to ensure operations yield positive net contribution margin.

Channel Mental Flows (Linear Diagnostic Chains)

END-TO-END REASONING

Mental Flow — Google Ads (Search & High Intent)

Impressions → *(filtered by)* **CTR** → yields **Clicks** → at a unit cost of **CPC** → *(converted on LP via)* **CVR** → producing **Conversions** → evaluated against unit acquisition cost **CPA** *(CPC ÷ CVR)* → generating **Revenue** / **AOV** → calculating ad efficiency **ROAS** *(AOV ÷ CPA)* → subtracted by **COGS** → determining product **Gross Margin %** → establishing **Break-even ROAS** → resulting in net cash flow **Contribution** / **Profit**.

Mental Flow — Meta Ads (Social & Interruption Discovery)

Reach → *(multiplied by repetition)* **Frequency** → generates total **Impressions** → filtered by hook/creative via **CTR** / **Link CTR** / **Outbound CTR** → driving outbound **Clicks** → purchased at auction unit rate **CPC** → converted through funnel via **CVR** → completing **Conversions** / **Results** → measured at unit acquisition rate **CPA** / **Cost per Result** → monetized via **Revenue** / **AOV** → determining scaling efficiency **ROAS** → factoring manufacturing / sourcing **COGS** → defining commercial **Gross Margin %** → benchmarked against **Break-even ROAS** → closing at net financial **Contribution** / **Profit**.

DIAGNOSTIC PLAYBOOK • PART 02

Upper Funnel & Traffic Diagnostics

Golden Rule: Observe Movement → Use Formula → Identify Possible Drivers → Investigate Drivers.

1. Impressions Changes

AUCTION & SCALE

▼ IMPRESSIONS DECREASED (↓) — CHECK:

- **Budget / Spend:** Daily caps hit, pacing throttled, or billing paused.
- **Delivery & Bidding:** Target bid cap set too low; outbid in auction.
- **Targeting / Audience:** Overly tight segment, extreme exclusions, overlap.
- **Ad Eligibility:** Policy issues, review delay, ad quality score penalty.

▲ IMPRESSIONS INCREASED (↑) — CHECK:

- Budget increased, audience broadened, or auction competition became favorable.

2. CTR (Click-Through Rate)

CREATIVE RESONANCE

```
CTR = (Clicks ÷ Impressions) × 100
```

▼ CTR DECREASED (↓) — CHECK:

- **Creative Fatigue:** Ad saturated, audience exhausted with old hook.
- **Ad Relevance:** Weak value hook, generic visuals, audience mismatch.
- **Audience Fit:** Algorithms pushing into broader, unengaged cohorts.
- **Placements:** Inventory shifting into passive or low-intent feeds.

▲ CTR INCREASED (↑) — CHECK:

- Fresh winning creative, tighter message-market fit, or high-intent placements.

3. Clicks Changes

TRAFFIC VOLUME

```
Clicks = Impressions × CTR
```

▼ CLICKS DECREASED (↓) — CHECK:

- **Formula Drivers:** Impressions ↓ OR CTR ↓ (or both falling together).
- Determine whether it is an *auction volume problem* (Impressions) or an *engagement problem* (CTR).

▲ CLICKS INCREASED (↑) — CHECK:

- **Formula Drivers:** Impressions ↑ OR CTR ↑ (or both). Cross-check LP bounce rate to ensure real visitor quality.

4. CPC (Cost Per Click)

CLICK EFFICIENCY

```
CPC = Spend ÷ Clicks
```

▲ CPC INCREASED (↑) — CHECK:

- **Direct Drivers:** Spend ↑ OR Clicks ↓ (or both).
- **Drill Down:** If Clicks ↓ → is CTR ↓ (fatigue) or Impressions ↓ (auction)?
- Examine rising CPM auction costs, aggressive manual bids, or degraded relevance.

▼ CPC DECREASED (↓) — CHECK:

- Spend ↓ OR Clicks ↑. Favorable bidding auction, cheaper placements, or surging CTR.

5. CVR (Conversion Rate)

LANDING PAGE / FUNNEL

```
CVR = (Conversions ÷ Clicks) × 100
```

▼ CVR DECREASED (↓) — CHECK:

- **Traffic Intent:** Irrelevant keyword searches or misleading ad message.
- **Landing Page UX:** Slow page load, broken mobile layout, form bugs.
- **Offer Friction:** Pricing friction, sudden shipping fees, lack of social proof.
- **Tracking / Tech:** Payment gateway failure, broken scripts, cookie tag loss.

▲ CVR INCREASED (↑) — CHECK:

- Conversions ↑ OR Clicks ↓. High intent buyers, streamlined 1-page checkout, clear offer.

6. Conversions Changes

OUTCOME VOLUME

```
Conversions = Clicks × CVR
```

▼ CONVERSIONS DECREASED (↓) — CHECK:

- **Formula Drivers:** Clicks ↓ OR CVR ↓ (or both).
- Identify whether the bottleneck sits in marketing traffic generation (Clicks) or on-site checkout funnel (CVR).

▲ CONVERSIONS INCREASED (↑) — CHECK:

- Clicks ↑ OR CVR ↑ (or both). Validate CPA & profit margins to maintain scaling health.

DIAGNOSTIC PLAYBOOK • PART 03

Cost, Lead & Auction Diagnostics

Deep dive into unit acquisition cost, lead capture, auction pressure, and basket sizes.

7. CPA (Cost Per Acquisition)

UNIT EFFICIENCY

```
CPA = CPC ÷ CVR = Spend ÷ Conversions
```

▲ **CPA INCREASED (↑) — CHECK:**

- **Core Equation:** CPC ↑ OR CVR ↓ (or both).
- **If CPC ↑:** Spend ↑ OR Clicks ↓ → Check auction bidding, fatigue, CPM inflation.
- **If CVR ↓:** Conversions ↓ OR Clicks ↑ → Check LP latency, checkout friction, tracking drop.

▼ **CPA DECREASED (↓) — CHECK:**

- CPC ↓ OR CVR ↑ (or both). Compounding efficiency gains across traffic & funnel.

8. CPL (Cost Per Lead)

LEAD EFFICIENCY

```
CPL = Spend ÷ Leads
```

▲ **CPL INCREASED (↑) — CHECK:**

- **Direct Drivers:** Spend ↑ OR Leads ↓ (or both).
- **Underlying Factors:** Rising CPCs, lower traffic, or drop in landing page form fills.
- **Quality Check:** Evaluate lead qualification fields vs form friction and bot spam.

▼ **CPL DECREASED (↓) — CHECK:**

- Spend ↓ OR Leads ↑. Streamlined forms, native lead ads, stronger value magnet.

9. CPM (Cost Per Mille)

AUCTION PRESSURE

```
CPM = (Spend ÷ Impressions) × 1,000
```

▲ **CPM INCREASED (↑) — CHECK:**

- **Auction & Seasonality:** Q4/holiday spikes, major competitor aggressive bids.
- **Audience:** Highly constrained audience sizes, narrow geo-targeting, heavy overlap.
- **Placements & Quality:** Constrained to expensive inventory, low ad quality score.

▼ **CPM DECREASED (↓) — CHECK:**

- Spend ↓ OR Impressions ↑. Broader audience cohorts, off-peak delivery, placement liquidity.

10. AOV (Average Order Value)

BASKET ECONOMICS

```
AOV = Revenue ÷ Orders
```

▼ **AOV DECREASED (↓) — CHECK:**

- **Product Mix:** Traffic skewing heavily to low-ticket introductory items.
- **Pricing & Promos:** Excessive coupon discounting cannibalizing basket revenue.
- **Upsell Performance:** Broken post-purchase cross-sell or bundle modules.
- **Customer Cohort:** Increased share of bargain hunters or price-sensitive users.

▲ **AOV INCREASED (↑) — CHECK:**

- Revenue ↑ OR Orders ↓. Bundles, minimum spend free-shipping thresholds, upsells.

11. ROAS (Return On Ad Spend)

EFFICIENCY MULTIPLIER

```
ROAS = Revenue ÷ Spend = AOV ÷ CPA
```

▼ **ROAS DECREASED (↓) — CHECK:**

- **Primary Ratio:** AOV ↓ OR CPA ↑ (or both).
- If **CPA** ↑: Drill down into CPC (CPM, CTR) and CVR (Landing Page/Offer).
- If **AOV** ↓: Investigate cart size decay or heavy promo code dilution.

▲ **ROAS INCREASED (↑) — CHECK:**

- AOV ↑ OR CPA ↓ (or both). Compounded gains from targeting and high-ticket baskets.

12. Revenue Changes

TOPLINE SCALE

```
Revenue = Conversions × AOV
```

▼ **REVENUE DECREASED (↓) — CHECK:**

- **Formula Drivers:** Conversions ↓ OR AOV ↓ (or both).
- Check macro demand patterns, tracking outages, or high-velocity inventory stock-outs.

▲ **REVENUE INCREASED (↑) — CHECK:**

- Conversions ↑ OR AOV ↑ (or both). Successful channel scaling with healthy unit economics.

DIAGNOSTIC PLAYBOOK • PART 04

Delivery, Saturation & Profitability

Meta platform delivery mechanics, creative fatigue, and unit economics profitability.

13. Spend Changes

PACING & BUDGET

▲ **SPEND INCREASED (↑) — CHECK:**

- Manual budget increase or automated scaling rules triggered.
- Delivery expansion into higher CPM auctions or competitive bid pools.
- Bid cap loosening causing aggressive clearing bid prices.

▼ **SPEND DECREASED (↓) — CHECK:**

- Budget cuts, billing threshold pauses, ad disapproval, tight bid caps.

14. Frequency Changes (Meta)

SATURATION ALERT

```
Frequency = Impressions ÷ Reach
```

▲ **FREQUENCY ↑ + CTR ↓ + CVR ↓ (CRITICAL ALERT):**

- **Audience Saturation** / **Fatigue:** Same audience has repeatedly seen creatives.
- **Remedy:** Refresh creative angles immediately, widen audience, exclude past 30-day purchasers.

▼ **FREQUENCY DECREASED (↓) — CHECK:**

- Reach ↑ OR Impressions ↓. Broadening audience or scaling fresh cohorts.

15. Reach Changes (Meta)

UNIQUE EXPOSURE

```
Reach = Impressions ÷ Frequency
```

▼ **REACH DECREASED (↓) — CHECK:**

- Impressions ↓ OR Frequency ↑ (or both).
- Audience pool constrained by excessive exclusion filters or bid restrictions.
- Ad delivery trapped in narrow retargeting loops rather than top-of-funnel prospecting.

▲ **REACH INCREASED (↑) — CHECK:**

- Targeting widened, lookalike expansion, budget scaled into fresh cohorts.

16. Gross Margin % Changes

UNIT PROFITABILITY

```
Gross Margin % = (Gross Profit ÷ Revenue) × 100
```

▼ **GROSS MARGIN DECREASED (↓) — CHECK:**

- COGS ↑ (supply cost increase, tariffs, higher fulfillment costs).
- Revenue per unit ↓ (deep promotional discounts, markdown cannibalization).
- **Direct Impact:** Gross Margin ↓ directly triggers *Break-even ROAS* ↑.

▲ **GROSS MARGIN INCREASED (↑) — CHECK:**

- COGS ↓ or price optimization. Allows profitable scaling at lower ROAS targets.

17. Break-even ROAS Changes

BENCHMARK

```
Break-even ROAS = 1 ÷ Gross Margin %
```

STRATEGIC FRAMEWORK:

- **Gross Margin** ↓ → Break-even ROAS ↑ (requires higher efficiency to survive).
- **Gross Margin** ↑ → Break-even ROAS ↓ (enables aggressive scaling leeway).
- **Rule:** If Actual ROAS < Break-even ROAS, every ad dollar burns capital.

18. Contribution / Profit

NET PROFITABILITY

```
Profit = Revenue − COGS − Ad Spend − Variable Costs
```

▼ **PROFIT DECREASED (↓) — CHECK:**

- Revenue ↓, COGS ↑, Ad Spend ↑, or Variable Costs ↑ (or combination).
- Check whether scaling ad spend triggered diminishing marginal returns where marginal CPA outpaced marginal gross profit.

▲ **PROFIT INCREASED (↑) — CHECK:**

- Revenue expanding faster than total marginal COGS and ad spend combined.